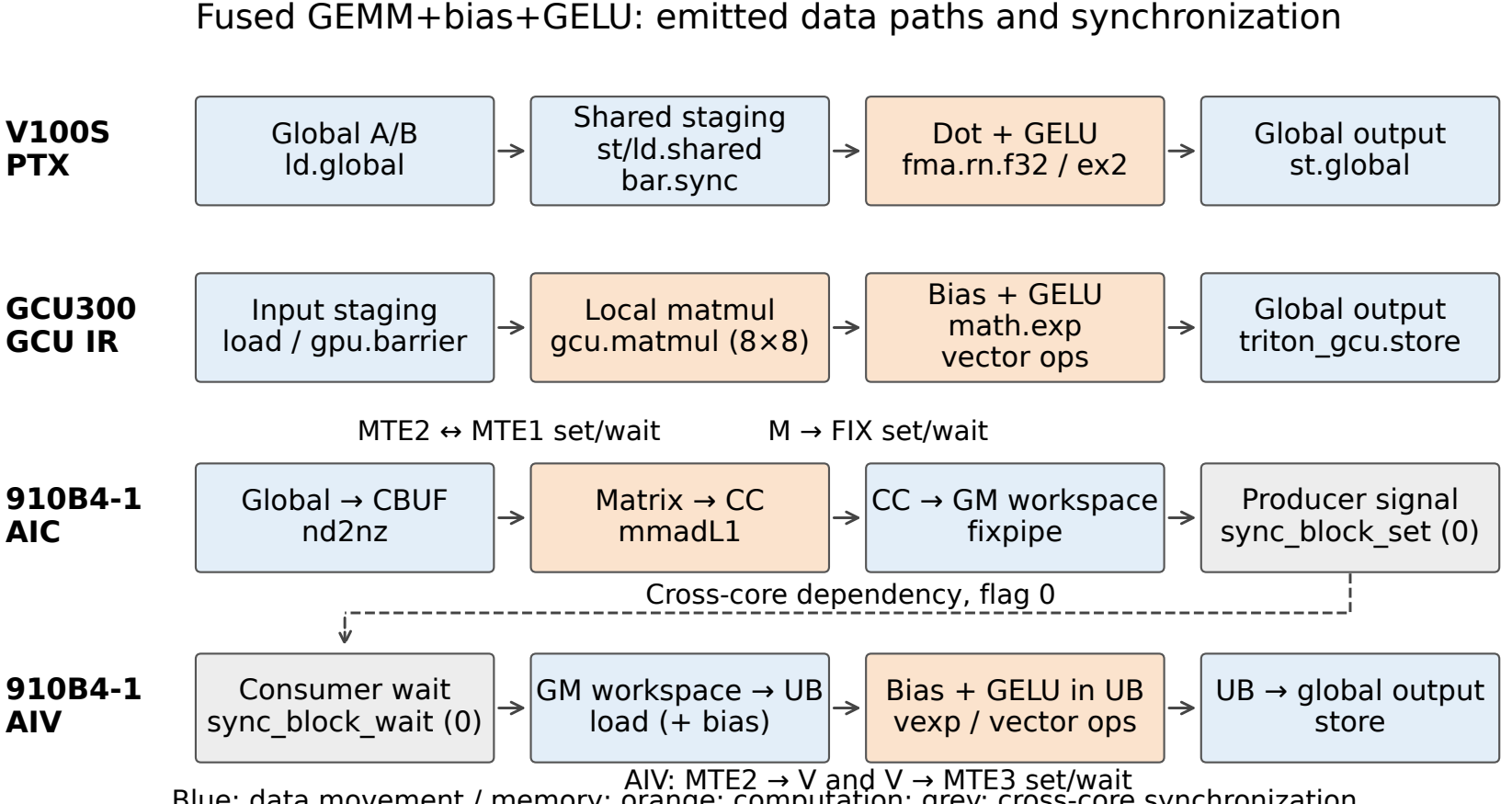
Fused GEMM+bias+GELU: emitted data paths and synchronization
V100S PTX
Global A/B ld.global
Shared staging st/ld.shared bar.sync
Dot + GELU fma.rn.f32 / ex2
Global output st.global
GCU300 GCU IR
Input staging load / gpu.barrier
Local matmul gcu.matmul (8×8)
Bias + GELU math.exp vector ops
Global output triton_gcu.store
MTE2 ↔ MTE1 set/wait
M → FIX set/wait
910B4-1 AIC
Global → CBUF nd2nz
Matrix → CC mmadL1
CC → GM workspace fixpipe
Producer signal sync_block_set (0)
Cross-core dependency, flag 0
910B4-1 AIV
Consumer wait sync_block_wait (0)
GM workspace → UB load (+ bias)
Bias + GELU in UB vexp / vector ops
UB → global output store
AIV: MTE2 → V and V → MTE3 set/wait
Blue: data movement / memory; orange: computation; grey: cross-core synchronization.
Selected emitted dependencies; box positions/widths carry no timing or overlap information.